MTH 202 Chapter 2: Vector-Valued Functions

2.6 Motion Along a Curve Worksheet

1. Find the velocity and position vectors of a particle whose acceleration is given by $\mathbf{a}(t) = \langle 2t, \sin(t), \cos(2t) \rangle$ with initial velocity $\mathbf{v}(0) = \mathbf{i}$ and initial position $\mathbf{r}(0) = \mathbf{j}$.

2. The position function of a particle is given by $\mathbf{r}(t) = \langle t^2 - 9t + 12, 4t + 5, 3t + 8 \rangle$. At what time is the particle travelling at its minimum speed?

3. Suppose a projectile is fired with an initial speed of 60 m/s at an angle of $45°$ from the initial position of of $\langle 1,1 \rangle$.

a) Write a vector function which will compute the position of the projectile at any time *t*.

b) At what time will the projectile hit the ground?

c) At what time will the projectile reach its maximum height?

d) Estimate the maximum height obtained by the projectile (round your answer to 2 decimals).

e) At what time will the projectile be 100 meters downrange?

f) Estimate the height of the projectile when it is 200 meters downrange (round your answer to 2 decimals).

g) Suppose we could change the angle at which the projectile is fired. What angle could we fire the projectile at so that it would be 100 meters downrange after 10 seconds?

4. Suppose that the velocity and acceleration vectors are perpendicular for the given curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. Then what can you say about the speed of this curve?